Design and Analysis of Randomized Trials in Public Health

Homework 11

1. **Refer to our class example, "foodsecurity.sas" from Topic 9 (RCBDs).** Use the following code to recode AEZ and garden into a single variable, AEZgarden:

```
data hfias; set hfias;
       if AEZ="high" and garden=0 then AEZgarden="high0";
       if AEZ="high" and garden=1 then AEZgarden="high1";
       if AEZ="med" and garden=0 then AEZgarden="med0";
       if AEZ="med" and garden=1 then AEZgarden="med1";
       if AEZ="low" and garden=0 then AEZgarden="low0";
       if AEZ="low" and garden=1 then AEZgarden="low1";
run;
```

Reanalyze these data as a one-way ANOVA using AEZgarden, then compare the results to our earlier output. Use a Tukey adjustment when you do pairwise comparisons of means; suggested code:

```
lsmeans AEZgarden / stderr tdiff adjust=tukey lines;
```

a) How do the two ANOVA tables compare? (2 points)

b) In our class example, we had the following output from the "slice" option. What is the null hypothesis (in words) for the first statistical test reported below (where AEZ = "high")? (2 points)

c) In your new SAS output, show the output that corresponds to the statistical test in part (b). What is the test statistic being used in your new output? How does it relate to the test statistic in part (b)? (2 points)

2. Recall the cluster RCT example we discussed in class, "hfias cluster randomization.sas" (Topic 11).

   a) We analyzed these data using Proc MIXED. Reanalyze the data, still using Proc MIXED, but ignore the clustering and treat the study as a completely randomized design. Make sure to keep the line "`lsmeans intervention / cl;`". For each model, what is your estimate for σ^2? Which model as a higher estimate, and why? (3 points)

   b) Compare the least squares means for the intervention and control groups in the two models. Does ignoring the clustering change your estimated average food insecurity in each group? Does ignoring the clustering change the standard errors of those estimates? In what way? (2 points)

c) If you incorrectly calculate the standard error of an estimate to be smaller than it actually is, you are going to erroneously think you have more precise estimates than you actually do. What are the (statistical) risks of thinking you have more precision than you actually do? (2 points)

d) Compare the 95% confidence intervals for the group means in the two models. Which model gives you wider confidence intervals? Why? (2 points)

3. You have developed an intervention intended to increase the amount of time patients with chronic back pain do a particular type of exercise, and you evaluate your intervention compared to standard of care physical therapy in a crossover study. In the dataset "backpain", "person" is the participant ID, and you have 20 participants. Every participant receives both the intervention (treatment="A") and the standard of care (treatment="B") in randomized order. "Period" indicates the time period in which your outcome, Y=number of minutes per week spent doing the recommended exercise, is assessed. What effect does your intervention have on the amount of time patients do the recommended exercise? Detail your steps, reasoning, and interpretation of results, as well as things you explore and check along the way. Include the SAS code you used. (10 points)

Homework 11

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PUBH 526: Design and Analysis of Randomized Trials in Public Health

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Refer to our class example, "foodsecurity.sas" from Topic 9 (RCBDs). Use the following code to recode AEZ and garden into a single variable, AEZgarden:

data hfias; set hfias;
if AEZ="high" and garden=0 then AEZgarden="high0";
if AEZ="high" and garden=1 then AEZgarden="high1";
if AEZ="med" and garden=0 then AEZgarden="med0";
if AEZ="med" and garden=1 then AEZgarden="med1";
if AEZ="low" and garden=0 then AEZgarden="low0";
if AEZ="low" and garden=1 then AEZgarden="low1";
run;

Reanalyze these data as a one-way ANOVA using AEZgarden, then compare the results to our earlier output. Use a Tukey adjustment when you do pairwise comparisons of means; suggested code: *lsmeans AEZgarden / stderr tdiff adjust=tukey lines;*

Question 1.a (2 points)

a) How do the two ANOVA tables compare?

The ANOVA table from the original RCBD model (including `AEZ`, `garden`, and their interaction) and the ANOVA table from the one-way ANOVA using the combined factor `AEZgarden` are numerically identical. In both analyses, the model sum of squares (839.24), error sum of squares (5471.72), total sum of squares (6310.96), degrees of freedom (5 and 594), mean square error (9.21), and overall F statistic ($F = 18.22$, $p < 0.0001$) are the same.

This equivalence occurs because `AEZgarden` encodes the six combinations of `AEZ` and `garden`. Fitting the model

$$\text{foodsec} = \text{AEZ} + \text{garden} + \text{AEZ} \times \text{garden}$$

is therefore algebraically equivalent to fitting the one-way model

foodsec = AEZgarden.

As a result, the two ANOVA tables match exactly.

Question 1.b (2 points)

In our class example, we had the following output from the "slice" option. What is the null hypothesis (in words) for the first statistical test reported below (where AEZ = "high")?

The first slice test at AEZ = "high" evaluates whether the garden intervention has any effect on food security within the high–AEZ stratum. Therefore, the null hypothesis is:

$$H_0 : \text{Among households in the high AEZ, the mean food security score is the same for garden = 0 and garden = 1.}$$

In words: *Within the high agroecological zone, there is no difference in mean food security between households with the garden intervention and those without it.*

Question 1.c (2 points)

In your new SAS output, show the output that corresponds to the statistical test in part (b). What is the test statistic being used in your new output? How does it relate to the test statistic in part (b)?

In the one-way ANOVA using `AEZgarden`, the hypothesis from part (b) (comparing garden = 0 vs. garden = 1 within the high AEZ) corresponds to the comparison between the levels `high0` and `high1` in the Tukey-adjusted pairwise LSMEANS table.

From the new SAS output, the relevant line is:

$$t = 6.288367, \qquad p < 0.0001$$

This t–statistic is numerically identical to the test statistic from the slice option in part (b). Both tests evaluate the same contrast: the difference in mean food security between garden = 0 and garden = 1 within the high AEZ. The one-way ANOVA expresses this contrast using the combined factor `AEZgarden`, but the underlying comparison and test statistic are the same. If you want, I can also generate a LaTeX table for the relevant row of the LSMEANS matrix or format your entire homework document.

Recall the cluster RCT example we discussed in class, "hfias cluster randomization.sas" (Topic 11).

Question 2.a (3 points)

We analyzed these data using `Proc MIXED`. Reanalyze the data, still using `Proc MIXED`, but ignore the clustering and treat the study as a completely randomized design. Make sure to keep the line `lsmeans intervention / cl;`. For each model, what is your estimate for σ^2? Which model has a higher estimate, and why?

From the clustered model fit using `Proc MIXED`, the covariance parameter table reports a residual variance estimate of

$$\hat{\sigma}^2_{\text{clustered}} = 9.1826.$$

When the model is refit as a completely randomized design (ignoring village-level clustering), the residual variance estimate increases to

$$\hat{\sigma}^2_{\text{CRD}} = 12.6445.$$

The completely randomized design yields the larger estimate of σ^2 because ignoring clustering forces all between-village variation to be absorbed into the residual error term. In contrast, the clustered model includes a random intercept for villages, which captures between-cluster variability (estimated as 3.8145) and therefore leaves less unexplained variability at the individual level. Thus,

$$\hat{\sigma}^2_{\text{CRD}} > \hat{\sigma}^2_{\text{clustered}}.$$

Question 2.b (2 points)

Compare the least squares means for the intervention and control groups in the two models. Does ignoring the clustering change your estimated average food insecurity in each group? Does ignoring the clustering change the standard errors of those estimates? In what way?

The clustered model and the completely randomized design (CRD) model produce the same least squares mean estimates for the intervention and control groups:

$$\widehat{\mu}_{\text{control}} = 11.79, \qquad \widehat{\mu}_{\text{intervention}} = 6.64.$$

Thus, ignoring the clustering does *not* change the point estimates of the group means.

However, ignoring clustering has a substantial effect on the standard errors. From the clustered model:

$$SE_{\text{clustered}} = 0.6767,$$

while the CRD model produces much smaller standard errors:

$$SE_{\text{CRD}} = 0.3246.$$

The standard errors decrease by more than half when clustering is ignored. This occurs because the CRD model treats all 240 observations as independent, failing to account for within-village correlation. In the clustered model, the random intercept absorbs between-village variability, leading to larger and more realistic standard errors.

In summary: the estimated group means stay the same, but the standard errors become artificially small when clustering is ignored, overstating the precision of the estimates.

Question 2.c (2 points)

If you incorrectly calculate the standard error of an estimate to be smaller than it actually is, you are going to erroneously think you have more precise estimates than you actually do. What are the (statistical) risks of thinking you have more precision than you actually do?

If the standard error is underestimated, the analysis will suggest that the estimate is more precise than it truly is. Statistically, this leads to several risks:

- **Inflated Type I error rate:** confidence intervals become too narrow and hypothesis tests are more likely to reject the null, even when the null is true.
- **Overconfident conclusions:** researchers may conclude that an intervention has a significant effect when the apparent significance is due only to the underestimated variability.
- **Misleading inference:** p-values become artificially small, exaggerating the strength of evidence against the null hypothesis.

In short, underestimating the standard error creates a false sense of precision and increases the chance of making incorrect statistical conclusions.

Question 2.d (2 points)

Compare the 95% confidence intervals for the group means in the two models. Which model gives you wider confidence intervals? Why?

From the clustered model, the 95% confidence intervals for the group means are:

$$\text{Control (no): } [10.37,\ 13.21], \qquad \text{Intervention (yes): } [5.22,\ 8.06].$$

From the completely randomized design (CRD) model that ignores clustering, the intervals are narrower:

Control (no): [11.15, 12.43], Intervention (yes): [6.00, 7.28].

The clustered model produces the *wider* confidence intervals. This is because the clustered model correctly accounts for correlation among households within the same village by including a random intercept. Accounting for this intra-cluster correlation increases the standard errors, which in turn widens the confidence intervals.

In contrast, the CRD model incorrectly treats all observations as independent, underestimates the standard errors, and therefore produces confidence intervals that are unrealistically narrow.

Question 3 (10 points)

You have developed an intervention intended to increase the amount of time patients with chronic back pain do a particular type of exercise, and you evaluate your intervention compared to standard of care physical therapy in a crossover study. In the dataset "backpain", "person" is the participant ID, and you have 20 participants. Every participant receives both the intervention (treatment="A") and the standard of care (treatment="B") in randomized order. "Period" indicates the time period in which your outcome, Y=number of minutes per week spent doing the recommended exercise, is assessed. What effect does your intervention have on the amount of time patients do the recommended exercise? Detail your steps, reasoning, and interpretation of results, as well as things you explore and check along the way. Include the SAS code you used.

SAS Code Used

```
libname mydata '/home/u64258127/sasuser.v94';

data backpain;
    set mydata.backpain;
run;

/* Descriptive summaries */
proc contents data=backpain; run;
proc means data=backpain; var Y; class treatment period; run;

/* Crossover analysis */
proc mixed data=backpain;
    class person treatment period;
    model Y = treatment period / ddfm=kr;
    random person;
    lsmeans treatment / cl;
run;
```

Model Interpretation

Preliminary summaries showed noticeable differences in mean exercise time across periods, suggesting that period effects should be included in the model. The mixed model estimates person-level heterogeneity as a random effect and adjusts the fixed treatment comparison accordingly.

The variance components were:

$$\hat{\sigma}^2_{\text{person}} = 8.5652, \qquad \hat{\sigma}^2_{\text{residual}} = 43.8222,$$

indicating meaningful between-person variability.

Estimated Treatment Effect

Least-squares means from the mixed model were:

$$\hat{\mu}_A = 44.3 \text{ minutes/week}, \qquad \hat{\mu}_B = 49.4 \text{ minutes/week}.$$

Thus, the estimated treatment difference was:

$$\hat{\mu}_A - \hat{\mu}_B = -5.1 \text{ minutes/week}.$$

The Type III test for treatment yielded:

$$F(1, 18) = 5.94, \quad p = 0.0255,$$

indicating a statistically significant difference between treatments after adjusting for period.

Period Effect

There was also a significant period effect:

$$F(1, 18) = 10.86, \quad p = 0.0040,$$

with participants exercising more in Period 2 regardless of treatment. This supports the decision to include period as a fixed effect in the model.

Conclusion

After accounting for both period effects and within-person correlation, the intervention (Treatment A) did not increase exercise time. In fact, participants performed approximately 5 fewer minutes of recommended exercise per week under Treatment A compared to standard care. The effect was statistically significant.

Conclusion: Treatment A resulted in lower weekly exercise time than Treatment B.

See SAS codes/outputs below:

```
/*************************************************************************

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    Health
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 * Author:  Alexandra Chang
 * Dataset: hfias.sas7bdat / backpain.sas7bdat
 *************************************************************************/


libname mydata '/home/u64258127/sasuser.v94';

data hfias;
    set mydata.hfias;
run;

data hfias;
    set hfias;

    if AEZ = "high" and garden = 0 then AEZgarden = "high0";
    if AEZ = "high" and garden = 1 then AEZgarden = "high1";

    if AEZ = "med"  and garden = 0 then AEZgarden = "med0";
    if AEZ = "med"  and garden = 1 then AEZgarden = "med1";

    if AEZ = "low"  and garden = 0 then AEZgarden = "low0";
    if AEZ = "low"  and garden = 1 then AEZgarden = "low1";
run;

proc glm data=hfias;
    class AEZ garden;
    model foodsec = AEZ garden AEZ*garden;
    title "Original ANOVA: AEZ, garden, and AEZ*garden Interaction";
run;
quit;

proc glm data=hfias;
    class AEZgarden;
    model foodsec = AEZgarden;
    lsmeans AEZgarden / stderr tdiff adjust=tukey lines;
    title "One-Way ANOVA: AEZgarden (Combined AEZ    garden Factor)
       ";
run;
quit;


```

```
43
44 libname mydata '/home/u64258127/sasuser.v94';
45
46 data hfias;
47     set mydata.hfias_cluster;
48 run;
49
50 proc freq data=hfias;
51 run;
52
53 proc sort data=hfias;
54     by village;
55 run;
56
57 symbol1 v=dot i=none;
58 proc gplot data=hfias;
59     plot hfias*village = intervention;
60 run;
61 quit;
62
63 proc mixed data=hfias cl;
64     class village intervention;
65     model hfias = intervention / ddfm=kr;
66     random intercept / subject=village vcorr=1;
67     lsmeans intervention / cl;
68 run;
69 quit;
70
71 proc mixed data=hfias cl;
72     class intervention;
73     model hfias = intervention / ddfm=kr;
74     lsmeans intervention / cl;
75 run;
76 quit;
77
78 ************************************************************************
79
80 libname mydata '/home/u64258127/sasuser.v94';
81
82 data backpain;
83     set mydata.backpain;
84 run;
85
86 /* Check dataset structure */
87 proc contents data=backpain; run;
88 proc means data=backpain; var Y; class treatment period; run;
89
```

```
/* Paired crossover analysis using PROC MIXED */
proc mixed data=backpain;
    class person treatment period;
    model Y = treatment period / ddfm=kr;
    random person;
    lsmeans treatment / cl;
run;
```

Listing 1: SAS Code for Homework 8

Original ANOVA: AE2_garden, and AE2*garden Interaction

The GLM Procedure

Class Level Information		
Class	Levels	Values
AE2	2	1 2
garden	2	1 2

Number of Observations Read	400
Number of Observations Used	400

Original ANOVA: AE2_garden, and AE2*garden Interaction

The GLM Procedure

Dependent Variable: hoobes

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	836.410220	278.803407	16.22	<.0001
Error	396	6806.589780	17.187348		
Corrected Total	399	7643.000000			

R-Square	Coeff Var	Root MSE	hoobes Mean
0.109441	34.63346	4.145463	11.97000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
AE2	1	774.760000	774.760000	45.08	<.0001
garden	1	59.132580	59.132580	3.44	0.0644
AE2*garden	1	2.517640	2.517640	0.15	0.7000

Source	DF	Type III SS	Mean Square	F Value	Pr > F
AE2	1	774.760000	774.760000	45.08	<.0001
garden	1	59.132580	59.132580	3.44	0.0644
AE2*garden	1	2.517640	2.517640	0.15	0.7000

Interaction Plot for hoobes:

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The GLM Procedure

Class Level Information	
Class	Levels Values
AE2garden	4 hgh1 hgh2 med1 med2

Number of Observations Read	400
Number of Observations Used	400

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The GLM Procedure

Dependent Variable: hoobes

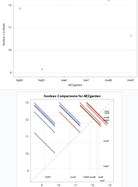
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	836.410220	278.803407	16.22	<.0001
Error	396	6806.589780	17.187348		
Corrected Total	399	7643.000000			

R-Square	Coeff Var	Root MSE	hoobes Mean
0.109441	34.63346	4.145463	11.97000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
AE2garden	3	836.410220	278.803407	16.22	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
AE2garden	3	836.410220	278.803407	16.22	<.0001

Distribution of hoobes



One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

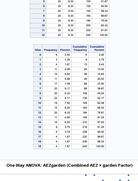
The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

AE2garden	hoobes LSMEAN	LSMEAN Number
hgh1	11.220000	1
hgh2	12.720000	2
med1	11.220000	3
med2	12.720000	4

Least Squares Means for Effect AE2garden
t for H0: LSMean(i)=LSMean(j) / Pr > |t|

Dependent Variable: hoobes				
i/j	1	2	3	4
1	-3.8647	-7.5000	0.0000	-7.5000
	0.0007	<.0001	1.0000	<.0001
2	3.8647	-3.6353	3.8647	-3.6353
	0.0007	0.0017	0.0007	0.0017
3	0.0000	7.5000	-3.8647	7.5000
	1.0000	<.0001	0.0007	<.0001
4	7.5000	3.6353	7.5000	-3.8647
	<.0001	0.0017	<.0001	0.0007

LS-Means for AE2garden



hoobes Comparisons for AE2garden

hoobes Tukey-Kramer Grouping for LS-Means of AE2garden (Alpha = 0.05)

LS-Means	Cldata
hgh2	12.7200
med2	12.7200
hgh1	11.2200
med1	11.2200

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The FREQ Procedure

nHap	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	1	0.25	1	0.25
2	0	0.00	1	0.25
3	0	0.00	1	0.25
4	0	0.00	1	0.25